

Final Path Experiment - Chronological Split Results

Date: 2026-06-08

Setup: fly_threshold_10km (10 km fly/stationary threshold)

Split: chronological 80/20 by target date (first 80% train, last 20% test; no shuffling)

k values: 7, 14, 30

Epochs: max 200, patience 30

Important Evaluation Note

The current training loop uses the chronological test partition for learning-rate scheduling, early stopping, and checkpoint selection. There is no separate validation set in this experiment.

With Weather

k=7

Model	Mean km	Median km	P90 km	Mig.50 mean km	Fly recall	Best epoch
direct_transformer_4l_k7	15.98	2.79	52.39	80.59		12
triline_lstm_4l_k7	47.87	2.26	52.02	106.33	0.693	3
triline_transformer_v2_k7	17.89	2.71	55.89	97.93	0.721	10
direct_mlp_sequence_k7	47.06	4.67	55.32	83.64		11

Winner by mean error: **direct_transformer_4l_k7** (15.98 km).

k=14

Model	Mean km	Median km	P90 km	Mig.50 mean km	Fly recall	Best epoch
triline_lstm_4l_k14	47.73	2.44	60.12	94.90	0.660	5
direct_transformer_4l_k14	17.08	4.43	56.60	86.62		4
triline_transformer_v2_k14	18.16	2.36	58.96	98.79	0.718	21
direct_mlp_sequence_k14	48.01	4.43	60.60	90.85		7

Winner by mean error: **triline_lstm_4l_k14** (17.73 km).

k=30

Model	Mean km	Median km	P90 km	Mig.50 mean km	Fly recall	Best epoch
direct_transformer_4l_k30	15.41	2.95	46.72	83.55		12
triline_transformer_v2_k30	15.05	2.28	45.46	98.99	0.728	21
triline_lstm_4l_k30	16.03	2.39	51.14	89.67	0.669	5
direct_mlp_sequence_k30	17.66	4.66	51.39	95.75		5

Winner by mean error: **direct_transformer_4l_k30** (15.41 km).

Without Weather

k=7

Model	Mean km	Median km	P90 km	Mig.50 mean km	Fly recall	Best epoch
triline_transformer_v2_k7	17.38	72.63	56.20	95.35	0.736	27
triline_lstm_4l_k7	17.97	2.37	58.29	100.39	0.727	7
direct_transformer_4l_k7	18.12	3.77	59.79	79.37		10
direct_mlp_sequence_k7	18.59	4.24	61.20	85.62		10

Winner by mean error: **triline_transformer_v2_k7** (17.38 km).

k=14

Model	Mean km	Median km	P90 km	Mig.50 mean km	Fly recall	Best epoch
triline_transformer_v2_k14	17.62	12.40	55.95	99.17	0.716	26
direct_transformer_4l_k14	18.03	13.25	59.64	81.79		14
triline_lstm_4l_k14	18.21	2.30	59.06	104.56	0.737	7
direct_mlp_sequence_k14	18.83	3.90	62.44	89.72		6

Winner by mean error: **triline_transformer_v2_k14** (17.62 km).

k=30

Model	Mean km	Median km	P90 km	Mig.50 mean km	Fly recall	Best epoch
triline_transformer_v2_k30	15.76	3.31	49.46	91.97	0.711	19
triline_lstm_4l_k30	15.93	2.38	49.55	91.30	0.663	11
direct_transformer_4l_k30	16.57	3.76	55.86	82.70		11
direct_mlp_sequence_k30	18.43	5.26	50.98	90.07		5

Winner by mean error: **triline_transformer_v2_k30** (15.76 km).

Weather Impact

Weather effect is reported as **weather - no weather**; negative values mean weather improves mean error.

Model	k=7	k=14	k=30
Triline LSTM 4L	-0.13	-0.48	+0.16
Triline Transformer V2	+0.51	+0.54	+0.18

Weather now helps the LSTM at k=7 and k=14, slightly hurts it at k=30, and hurts Transformer mean error at all k values. Weather still improves some tail-error slices, including Transformer P90 at k=30.

Rollout Evaluation

Condition	Direct mean km	LSTM mean km	Transformer mean km	Persistence mean km
with_weather	316.03	136.61	144.11	388.62
without_weather	534.17	178.37	129.55	388.62

Rollout results are mixed: weather substantially improves Direct and LSTM rollouts but the no-weather Transformer has the lowest mean rollout error. All learned rollouts remain sensitive to compounding path-level drift.

Conclusions

1. The chronological split produces strong one-step performance, especially for long-context Transformer models.
2. With weather at k=30, Direct Transformer 4L has the best mean error (15.41 km) and Triline Transformer V2 has the best median/P90 among triline models.
3. Without weather, Triline Transformer V2 is the strongest model across k=7,14,30 by mean error.
4. Weather is useful but not uniformly: it improves Direct models strongly and helps some LSTM settings, while Transformer mean error increases slightly with weather.
5. Autoregressive rollout remains the harder setting because errors compound over full migration paths.